

Finiteness in Square Classes and Weighted Zero Density of Perfect Cuboids

Valery Asiryan

asiryanvalery@gmail.com

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Abstract

Let $M(N)$ be the number of similarity classes of positive rational perfect cuboids with ordered edges a, b, c , space diagonal g , and squarefree invariant $N = \text{sf}(abcg/2)$. We prove that $M(N)$ is finite for every positive squarefree N , without any restriction on the rank of the associated congruent-number elliptic curve. More precisely, $M(N) \leq C^{1+r_N}$ for an absolute constant $C > 1$, where r_N is the rank of $y^2 = x^3 - N^2x$. Using the established Paulsen–West rank obstruction and Smith’s results on quadratic twists, we then prove

$$\lim_{X \rightarrow \infty} \frac{1}{X} \sum_{\substack{N \leq X \\ N \text{ positive squarefree}}} M(N)^q = 0 \quad \text{for every fixed } q > 0.$$

Thus the result counts cuboid classes with their full multiplicities at each N . A quantitative decay estimate is also obtained. The geometric step excludes every positive-dimensional real abelian coset through the positive part of the compatibility surface; quantitative Mordell–Lang then gives the required bound. The argument is unconditional and uses the cuboid correspondence and rank obstruction as prior results.

Keywords: perfect cuboids; congruent numbers; elliptic curves; Mordell–Lang theorem; quadratic twists; zero density.

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1 Introduction and main results

A positive rational perfect cuboid consists of seven positive rational numbers satisfying

$$\begin{aligned} a^2 + b^2 &= d^2, & a^2 + c^2 &= e^2, & b^2 + c^2 &= f^2, \\ a^2 + b^2 + c^2 &= g^2. \end{aligned} \tag{1}$$

We keep the edges a, b, c ordered and identify two cuboids when all lengths differ by the same positive rational factor. Such similarity classes are equivalently primitive integral perfect cuboids with ordered edges.

For $u \in \mathbb{Q}_{>0}$, let $\text{sf}(u)$ be the unique positive squarefree integer representing its class in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$. Define

$$N = \text{sf}\left(\frac{abcg}{2}\right), \quad E_N : y^2 = x^3 - N^2x, \quad r_N = \text{rank } E_N(\mathbb{Q}). \quad (2)$$

The integer N is a similarity invariant: scaling the lengths by $v \in \mathbb{Q}_{>0}$ multiplies $abcg/2$ by the square v^4 .

Let $\mathcal{S}$ be the set of positive squarefree integers, and put

$$M(N) = \#\{\text{ordered similarity classes satisfying (1) with invariant } N\}. \quad (3)$$

The count concerns similarity classes; without this identification, any one rational cuboid would give infinitely many scaled copies with the same N .

Paulsen and West [1, Proposition 3.1] express the cuboid condition by three rational points on a congruent-number curve. Their Theorem 4.2 gives the rank obstruction

$$M(N) > 0 \implies r_N \geq 2. \quad (4)$$

We use these results to control the number of cuboids at a fixed N and then sum these numbers over the squarefree parameters.

Theorem 1 (Finiteness at every fixed parameter). *There is an absolute constant $C_0 > 1$ such that, for every $N \in \mathcal{S}$,*

$$M(N) \leq C_0^{1+r_N} < \infty. \quad (5)$$

There is no upper bound or other restriction on r_N in this statement. One admissible choice is

$$C_0 = 108^{180.4^{38}}. \quad (6)$$

Theorem 2 (Weighted zero density). *For every fixed real number $q > 0$,*

$$\lim_{X \rightarrow \infty} \frac{1}{X} \sum_{\substack{N \in \mathcal{S} \\ N \leq X}} M(N)^q = 0. \quad (7)$$

More precisely, there is an absolute constant $\gamma > 0$ such that for each fixed $q > 0$ there are positive constants K_q, D_q, X_q with

$$\sum_{\substack{N \in \mathcal{S} \\ N \leq X}} M(N)^q \leq K_q X \exp\left(-\gamma L(X)^{1/2} + D_q L(X)^{1/4}\right) \quad (X \geq X_q), \quad (8)$$

where $L(X) = \log \log \log X$.

All logarithms are natural, and expressions involving $L(X)$ are used only for sufficiently large X . Theorem 1 is geometric and applies uniformly across all ranks. Theorem 2 combines that bound with (4), the scarcity of twists of rank at least two, and exponential rank moments. The latter two inputs are due to Smith [2, 3]. The moment method is part of the

existing arithmetic framework; the specific ingredient here is the geometry of the positive cuboid configurations.

To make the consequence for the original objects explicit, let

$$\mathcal{C}(X) = \{\text{ordered cuboid similarity classes with } \text{sf}(abcg/2) \leq X\}.$$

The case $q = 1$ gives

$$\#\mathcal{C}(X) = \sum_{\substack{N \in \mathcal{S} \\ N \leq X}} M(N), \quad \lim_{X \rightarrow \infty} \frac{\#\mathcal{C}(X)}{X} = 0. \quad (9)$$

This is density with respect to the specified squarefree invariant. It is not a count ordered by maximum edge length or projective height.

2 The Paulsen–West model in positive coordinates

We record the coordinate normalization needed below, using the correspondence of Paulsen–West [1, Proposition 3.1]. Their parameters satisfy

$$N\eta_i^2 = t_i - t_i^3, \quad 0 < t_i < 1, \quad t_1 t_2 + t_1 t_3 + t_2 t_3 = 1. \quad (10)$$

For a cuboid with $abcg/2 = Ns^2$, $s > 0$, its half-angle parameters are

$$t_1 = \frac{ag}{de + bc}, \quad t_2 = \frac{bg}{df + ac}, \quad t_3 = \frac{cg}{ef + ab}. \quad (11)$$

The associated triangle has sides af, be, cd and area $abcg/2$, so the curve parameter in (10) is exactly the invariant (2).

For the counting argument we use points on E_N in the region $x > N$, $y > 0$. An explicit map from the cuboid is

$$\begin{aligned} x_1 &= N \frac{de + ag}{bc}, & y_1 &= \frac{ag}{s} x_1, \\ x_2 &= N \frac{df + bg}{ac}, & y_2 &= \frac{bg}{s} x_2, \\ x_3 &= N \frac{ef + cg}{ab}, & y_3 &= \frac{cg}{s} x_3. \end{aligned} \quad (12)$$

These are the usual congruent-number points obtained from the three right triangles

$$\frac{1}{s}(ag, bc, de), \quad \frac{1}{s}(bg, ac, df), \quad \frac{1}{s}(cg, ab, ef),$$

each of area N . Indeed, for a rational right triangle (U, V, H) with $UV = 2N$, the formulas

$$x = N \frac{H + U}{V}, \quad y = Ux \quad (13)$$

give $x > N$, $y > 0$, and $x^2 - N^2 = U^2 x$, hence $y^2 = x^3 - N^2 x$. The identity $U/(H + V) = (x - N)/(x + N)$ shows that

$$t_i = \frac{x_i - N}{x_i + N}. \quad (14)$$

In these coordinates, the compatibility equation in (10) becomes

$$F_N(x_1, x_2, x_3) = x_1x_2x_3 - N(x_1x_2 + x_1x_3 + x_2x_3) - N^2(x_1 + x_2 + x_3) + N^3 = 0. \quad (15)$$

For example, the exact expansion is

$$1 - \sum_{i < j} t_i t_j = \frac{-2F_N(x_1, x_2, x_3)}{(x_1 + N)(x_2 + N)(x_3 + N)}.$$

Let $S_N \subset E_N^3$ be the inverse image of the multihomogeneous equation (15) under the three degree-two maps $x : E_N \rightarrow \mathbb{P}^1$. Define

$$S_N^+(\mathbb{Q}) = \{(P_1, P_2, P_3) \in S_N(\mathbb{Q}) : x(P_i) > N, y(P_i) > 0 \text{ for all } i\}. \quad (16)$$

Proposition 1. *The map (12) induces an injection from the classes counted by $M(N)$ into $S_N^+(\mathbb{Q})$. In particular,*

$$M(N) \leq \#S_N^+(\mathbb{Q}). \quad (17)$$

Proof. The preceding formulas establish membership in $S_N^+(\mathbb{Q})$. Scaling all cuboid lengths by $v > 0$ scales s by v^2 and leaves the three points unchanged. Finally, the points determine the ordered edge ratios through

$$a : b : c = \frac{y_1}{x_1} : \frac{y_2}{x_2} : \frac{y_3}{x_3}.$$

These ratios determine the similarity class. This proves injectivity. $\square$

Only this injection is needed. The converse cuboid construction belongs to the established correspondence [1] and is not reproved here.

3 Geometry of the compatibility surface

3.1 Irreducibility and degree

Lemma 1. *For every $N \in \mathcal{S}$, the surface S_N is geometrically integral. If*

$$D_i = \text{pr}_i^{-1}(O), \quad \Theta = D_1 + D_2 + D_3, \quad \mathcal{L} = \mathcal{O}(3\Theta),$$

then

$$[S_N] = 2\Theta, \quad \Theta^3 = 6, \quad \deg_{\mathcal{L}}(S_N) = 108. \quad (18)$$

Proof. Over $\mathbb{C}$, the substitutions

$$x_i = Nu_i, \quad y_i = N\sqrt{N}v_i$$

identify all the surfaces with $S_1 \subset E_1^3$. Write the three abscissas as u, v, z . The base surface in $(\mathbb{P}^1)^3$ is the closure of the graph

$$z = \frac{A}{D}, \quad A = uv + u + v - 1, \quad D = uv - u - v - 1. \quad (19)$$

The polynomials A, D are coprime. Thus this base surface is integral and has function field $\mathbb{C}(u, v)$.

Its inverse image in E_1^3 is obtained generically by adjoining square roots of

$$f_1 = u(u^2 - 1), \quad f_2 = v(v^2 - 1), \quad f_3 = z(z^2 - 1).$$

Their square classes are independent. Along the irreducible divisor

$$A = (u + 1)(v + 1) - 2 = 0,$$

the functions f_1, f_2 have order zero and f_3 has order one. Any product $f_1^{\epsilon_1} f_2^{\epsilon_2} f_3^{\epsilon_3}$ which is a square, with $\epsilon_i \in \{0, 1\}$, must therefore have $\epsilon_3 = 0$. Valuations along $u = 0$ and $v = 0$ then give $\epsilon_1 = \epsilon_2 = 0$. The generic inverse image consequently has a field of functions of degree eight over $\mathbb{C}(u, v)$.

The map $E_1^3 \rightarrow (\mathbb{P}^1)^3$ is finite flat of degree eight. Its base change to the integral base surface remains finite flat. Its coordinate algebra over each affine open of the base surface embeds into its generic fiber, which is a field by the preceding calculation. Hence the entire inverse image is integral, including its boundary.

The base equation has multidegree $(1, 1, 1)$, and $x^* \mathcal{O}_{\mathbb{P}^1}(1) = \mathcal{O}_{E_N}(2\Theta)$. Therefore $[S_N] = 2\Theta$. Since $D_i^2 = 0$ and $D_1 D_2 D_3 = 1$, one has $\Theta^3 = 6$. Finally,

$$\deg_{\mathcal{L}}(S_N) = (2\Theta)(3\Theta)^2 = 108.$$

The line bundle $\mathcal{L}$ is very ample, being the exterior product of the degree-three embeddings of the three elliptic factors. $\square$

3.2 The real obstruction to abelian cosets

Lemma 2. *Let $p \in S_N(\mathbb{R})$ satisfy $p_i \neq O$ and $x(p_i) > N$ for $i = 1, 2, 3$. There is no positive-dimensional abelian subvariety $B \subset E_N^3$, defined over $\mathbb{R}$, such that*

$$p + B \subseteq S_N.$$

Proof. The identity component $E_N(\mathbb{R})^0$ is a circle, with abscissas in $[N, +\infty]$. Write $T_+ = (N, 0)$, and define on this circle

$$\theta(P) = 2 \arctan \frac{x(P) - N}{x(P) + N}, \quad \theta(O) = \frac{\pi}{2}. \quad (20)$$

This is continuous, takes values in $[0, \pi/2]$, and satisfies

$$\theta^{-1}(0) = \{T_+\}, \quad \theta^{-1}(\pi/2) = \{O\}. \quad (21)$$

On $S_N \cap (E_N(\mathbb{R})^0)^3$, including the boundary, one has

$$\theta(P_1) + \theta(P_2) + \theta(P_3) = \pi. \quad (22)$$

Indeed, the fractional linear coordinates in (14) extend to $t_i(O) = 1$ and lie in $[0, 1]$. The multihomogeneous equation remains exactly $\sum_{i < j} t_i t_j = 1$. For $\alpha_i = \arctan t_i \in [0, \pi/4]$,

$$\cos(\alpha_1 + \alpha_2 + \alpha_3) = \prod_i \cos \alpha_i \left(1 - \sum_{i < j} t_i t_j \right) = 0.$$

Since the sum of the α_i belongs to $[0, 3\pi/4]$, it equals $\pi/2$, proving (22).

Suppose now that $p+B \subseteq S_N$. The compact connected real Lie group $B(\mathbb{R})^0$ is a positive-dimensional torus. Choose a circle subgroup $C \subseteq B(\mathbb{R})^0$ and consider the translated circle $p+C$. Its three coordinate projections lie in $E_N(\mathbb{R})^0$. Each projection is either constant or a finite covering of that circle, and at least one is nonconstant.

If at least one projection is constant, let its coordinate index be i , so its angle is identically $c = \theta(p_i) \in (0, \pi/2)$. Since C is positive-dimensional, there is a nonconstant projection with index $j \neq i$. This projection covers the circle and reaches T_+ , where its angle is zero. At that point on $p+C$, the remaining coordinate $k \in \{1, 2, 3\} \setminus \{i, j\}$ would be forced by (22) to have angle $\pi - c - 0 = \pi - c > \pi/2$, which is impossible since θ takes values in $[0, \pi/2]$ (regardless of whether the k -th projection is constant or nonconstant).

It remains to consider three nonconstant projections, with covering degrees $n_1, n_2, n_3 \geq 1$. For the i th projection, let $A_i \subset C$ be the inverse image of T_+ and $Z_i \subset C$ the inverse image of O . Then

$$|A_i| = |Z_i| = n_i.$$

At a parameter in A_i , the other two angles must both equal $\pi/2$. Thus $A_i \subseteq Z_j \cap Z_k$ whenever $\{i, j, k\} = \{1, 2, 3\}$. The three sets A_i are pairwise disjoint, since two zero angles cannot occur in (22). Consequently,

$$n_1 \geq n_2 + n_3, \quad n_2 \geq n_1 + n_3, \quad n_3 \geq n_1 + n_2.$$

Their sum is incompatible with $n_1 + n_2 + n_3 > 0$. This contradiction proves the lemma. $\square$

4 Finiteness and a bound uniform in the parameter

4.1 One positive point on each geometric coset

The standard rational torsion calculation for congruent-number curves gives

$$E_N(\mathbb{Q})_{\text{tors}} = \{O, (0, 0), (N, 0), (-N, 0)\}; \quad (23)$$

see Koblitz [4]. We first note that the region $x > N$, $y > 0$ contains at most one point of each rational torsion orbit. For $P = (x, y)$ in that region, the nonzero torsion translations satisfy

$$\begin{aligned} x(P + (0, 0)) &= -N^2/x < 0, \\ x(P + (-N, 0)) &= -N(x - N)/(x + N) < 0, \\ x(P + (N, 0)) &= N(x + N)/(x - N) > N, \\ y(P + (N, 0)) &= -2N^2y/(x - N)^2 < 0. \end{aligned} \quad (24)$$

Thus no nonzero rational torsion translation preserves this region.

Lemma 3. *For any translate $C = t + B \subseteq S_N$ of an abelian subvariety of E_N^3 over $\overline{\mathbb{Q}}$,*

$$\#(C \cap S_N^+(\mathbb{Q})) \leq 1. \quad (25)$$

Proof. There is nothing to prove if the intersection is empty. Otherwise choose p in it, so $C = p + B$. Put $\Gamma = E_N(\mathbb{Q})^3$. Then

$$C \cap \Gamma = p + (B \cap \Gamma).$$

Let H be the Zariski closure of the subgroup $B \cap \Gamma$. It is an algebraic subgroup of E_N^3 . Because its defining dense set consists of rational points, it is defined over $\mathbb{Q}$. If $B \cap \Gamma$ were infinite, the identity component H^0 would be positive-dimensional. Since $H \subseteq B$, the inclusion

$$p + H^0 \subseteq S_N$$

would contradict Lemma 2.

Therefore $B \cap \Gamma$ is finite and consists of torsion points. Any two points of $C \cap S_N^+(\mathbb{Q})$ differ coordinatewise by rational torsion. Equations (23)–(24) show that they are equal. $\square$

The argument includes cosets defined only over $\overline{\mathbb{Q}}$. It is not necessary to assume that the original B descends to $\mathbb{R}$; the rational points produce the subgroup H^0 to which the real lemma applies.

4.2 Quantitative Mordell–Lang

We use the following result of David, Nakamaye, and Philippon. Let $g \geq 2$, let E be an elliptic curve over $\overline{\mathbb{Q}}$, let $V \subseteq E^g$ be a subvariety, and let $\Gamma \subseteq E^g(\overline{\mathbb{Q}})$ be a subgroup of finite rank r . For the polarization

$$\mathcal{L} = \bigotimes_{i=1}^g \text{pr}_i^* \mathcal{O}_E(3O),$$

the set $V \cap \Gamma$ is covered by translates of abelian subvarieties contained in V , whose number is at most

$$\left(\deg_{\mathcal{L}} V\right)^{60(r+1)(g+1)^4(\dim V+1)^2+2}. \quad (26)$$

This is [5, Theorem 1.11].

Proof of Theorem 1. Apply (26) with $E = E_N$, $g = 3$, $V = S_N$, and $\Gamma = E_N(\mathbb{Q})^3$. By Mordell–Weil, Γ has finite rank $3r_N$. Lemma 1 gives $\dim S_N = 2$ and $\deg_{\mathcal{L}} S_N = 108$, so the number of covering cosets is at most

$$108^{60(3r_N+1)4^{38}}.$$

Each coset contributes at most one point of $S_N^+(\mathbb{Q})$ by Lemma 3. Proposition 1 therefore gives

$$M(N) \leq \#S_N^+(\mathbb{Q}) \leq 108^{60(3r_N+1)4^{38}} \leq \left(108^{180 \cdot 4^{38}}\right)^{1+r_N}. \quad (27)$$

This proves the theorem for every N , with no rank restriction. The rank obstruction (4) has not been used in this proof. $\square$

For finiteness alone, ordinary Mordell–Lang and Lemma 3 suffice. The quantitative form is needed to control the sum of the multiplicities $M(N)$ as N varies. The constant in (6) is an admissible theoretical bound; no practical enumeration threshold is asserted.

5 The rank estimates used in the density argument

Let

$$\mathcal{R}(X) = \{N \in \mathcal{S} : N \leq X, r_N \geq 2\}, \quad L(X) = \log \log \log X.$$

We need two consequences of Smith’s work, with their precise parameter ranges.

Proposition 2 (Rank statistics). *There exist positive constants $A_1, A_2, \gamma, \beta, \kappa$ and X_0 such that, for every $X \geq X_0$,*

$$\#\mathcal{R}(X) \leq A_1 X \exp\left(-\gamma \sqrt{L(X)}\right), \quad (28)$$

and, uniformly for real m satisfying $1 \leq m < \kappa L(X)$,

$$\sum_{\substack{N \in \mathcal{S} \\ N \leq X}} e^{mr_N} \leq A_2 X e^{\beta m^2}. \quad (29)$$

Proof. For (28), apply Smith [2, Assumption 1.1(3), Theorem 1.2, Remark 1.3] to $E_1 : y^2 = x(x-1)(x+1)$. It has full rational two-torsion and no rational cyclic isogeny of degree four. The latter condition follows, for example, from the root-difference criterion in [3, Example 3.1, equation (3.1)]: the three relevant products are $-1, 2, 2$, none a rational square. Smith’s bound counts all nonzero integer twist parameters with 2^∞ -Selmer corank at least two. Restricting to positive squarefree parameters and using

$$r_N \leq \text{corank}_{\mathbb{Z}_2} \text{Sel}_{2^\infty}(E_N/\mathbb{Q})$$

gives (28). This inequality does not require finiteness of the Shafarevich–Tate group.

For (29), use [3, Theorem 1.1] with the fixed abelian variety $E_1/\mathbb{Q}$ and quadratic extensions of $\mathbb{Q}$. In the notation of that theorem, if $\mathcal{K}(H)$ denotes the quadratic fields of absolute discriminant at most H , then

$$\sum_{K \in \mathcal{K}(H)} e^{m \text{rank } E_1(K)} \leq e^{Cm^2} \#\mathcal{K}(H) \quad (30)$$

for $H > C$ and $1 \leq m < c \log \log \log H$, with constants depending only on E_1 .

For $N > 1$, take $K_N = \mathbb{Q}(\sqrt{N})$. Distinct squarefree N give distinct fields, and

$$|\text{Disc } K_N| \leq 4N, \quad r_N \leq \text{rank } E_1(K_N).$$

The rank inequality follows from the isomorphism

$$E_N \longrightarrow E_1, \quad (x, y) \longmapsto \left(\frac{x}{N}, \frac{y}{N\sqrt{N}} \right)$$

over K_N . Since $\#\mathcal{K}(4X) = O(X)$, inequality (30) implies (29) after adjusting the constants. The single $N = 1$ term can be absorbed uniformly for $m \geq 1$, since its rank is fixed and $m \leq m^2$. Decreasing κ if necessary gives the stated common range in terms of $L(X)$. $\square$

6 Proof of weighted zero density

Proof of Theorem 2. Fix $q > 0$ and put

$$\alpha = q \log C_0 > 0, \quad L = L(X).$$

By (4), every nonzero summand comes from $\mathcal{R}(X)$. By Theorem 1,

$$M(N)^q \leq e^{\alpha(1+r_N)}.$$

For any real $p > 1$ such that $1 \leq \alpha p < \kappa L$, Hölder's inequality and Proposition 2 give

$$\begin{aligned} \sum_{\substack{N \in \mathcal{S} \\ N \leq X}} M(N)^q &\leq e^\alpha (\#\mathcal{R}(X))^{1-1/p} \left(\sum_{\substack{N \in \mathcal{S} \\ N \leq X}} e^{\alpha p r_N} \right)^{1/p} \\ &\leq e^\alpha (A_1 X e^{-\gamma \sqrt{L}})^{1-1/p} (A_2 X e^{\beta \alpha^2 p^2})^{1/p} \\ &\leq e^\alpha A_3 X \exp \left(-\gamma \sqrt{L} + \frac{\gamma \sqrt{L}}{p} + \beta \alpha^2 p \right), \end{aligned} \quad (31)$$

where $A_3 = \max\{1, A_1, A_2\}$ is independent of p and X . If $\mathcal{R}(X)$ is empty, the desired bound is immediate and the same displayed upper bound still holds.

Choose

$$p = \frac{\sqrt{\gamma}}{\alpha \sqrt{\beta}} L^{1/4}. \quad (32)$$

Because q is fixed, sufficiently large X ensures $p > 1$ and

$$1 \leq \alpha p = \sqrt{\gamma/\beta} L^{1/4} < \kappa L.$$

Thus this choice lies in the uniform moment range. Substitution into (31) yields

$$\sum_{\substack{N \in \mathcal{S} \\ N \leq X}} M(N)^q \leq e^\alpha A_3 X \exp \left(-\gamma L^{1/2} + 2\alpha \sqrt{\beta \gamma} L^{1/4} \right). \quad (33)$$

This is (8), with

$$K_q = e^\alpha A_3, \quad D_q = 2q(\log C_0) \sqrt{\beta \gamma}.$$

Finally, $L(X) \rightarrow \infty$ and

$$-\gamma L(X)^{1/2} + D_q L(X)^{1/4} \longrightarrow -\infty.$$

Dividing (8) by X therefore gives the explicit limit

$$0 \leq \frac{1}{X} \sum_{\substack{N \in \mathcal{S} \\ N \leq X}} M(N)^q \leq K_q \exp \left(-\gamma L(X)^{1/2} + D_q L(X)^{1/4} \right) \longrightarrow 0. \quad (34)$$

The sum includes all squarefree N and all associated ranks. □

Corollary 1. For fixed $q > 0$ and $0 < \varepsilon < \gamma$,

$$\sum_{\substack{N \in \mathcal{S} \\ N \leq X}} M(N)^q \ll_{q,\varepsilon} X \exp \left(-(\gamma - \varepsilon) \sqrt{\log \log \log X} \right).$$

Proof. For sufficiently large X , the positive term $D_q L(X)^{1/4}$ in (8) is at most $\varepsilon L(X)^{1/2}$. □

7 Interpretation for the cuboid problem

Let $Q(X) = \#\{N \in \mathcal{S} : N \leq X\}$. The elementary squarefree counting formula is

$$Q(X) = \frac{6}{\pi^2}X + O(\sqrt{X}). \quad (35)$$

For completeness, the identity $\mu(n)^2 = \sum_{d^2|n} \mu(d)$ gives

$$Q(X) = \sum_{d \leq \sqrt{X}} \mu(d) \left\lfloor \frac{X}{d^2} \right\rfloor = X \sum_{d=1}^{\infty} \frac{\mu(d)}{d^2} + O(\sqrt{X}),$$

and the infinite sum equals $1/\zeta(2) = 6/\pi^2$.

Corollary 2. *For every fixed $q > 0$,*

$$\frac{1}{Q(X)} \sum_{\substack{N \in \mathcal{S} \\ N \leq X}} M(N)^q \longrightarrow 0. \quad (36)$$

In particular, the squarefree parameters admitting any perfect cuboid have relative density zero:

$$\frac{\#\{N \in \mathcal{S} : N \leq X, M(N) > 0\}}{Q(X)} \longrightarrow 0. \quad (37)$$

Proof. Equation (36) follows from (7) and (35). Since $M(N)$ is a nonnegative integer, $\mathbf{1}_{M(N)>0} \leq M(N)^q$, and (37) follows. $\square$

Equivalently, if N is chosen uniformly from the squarefree integers up to X , every fixed positive moment of the number of cuboid classes tends to zero. This controls the multiplicity at each parameter, not only the size of the set of admissible parameters. For $T \geq 1$, the same bound gives

$$\#\{N \in \mathcal{S} : N \leq X, M(N) \geq T\} \leq \frac{K_q X}{T^q} \exp\left(-\gamma L(X)^{1/2} + D_q L(X)^{1/4}\right) \quad (38)$$

whenever $X \geq X_q$. Here q is fixed before X tends to infinity.

If edge order is disregarded, the corresponding count is $M(N)/6$. Indeed, two equal positive rational edges would give an irrational face diagonal, so the three edges are distinct and all six permutations represent different ordered classes. Consequently all finiteness and density statements hold for unordered cuboids as well.

8 Conclusions

For every positive squarefree N , the number of positive rational perfect cuboids up to similarity is finite, regardless of the rank of E_N . The proof also gives the uniform estimate $M(N) \leq C_0^{1+r_N}$. Its essential geometric step is that no positive-dimensional real abelian coset contained in S_N can pass through its positive region; rational differences then reduce every geometric coset to at most one positive rational point.

Combining this estimate with the established Paulsen–West rank obstruction and Smith’s rank statistics proves that the total number of cuboid classes with invariant at most X is $o(X)$. More strongly, every fixed positive moment satisfies (7), with the quantitative decay in (8). The cuboid correspondence, the rank obstruction, quantitative Mordell–Lang, and the distributional results are used with their original attribution.

These conclusions leave open whether any perfect cuboid exists, and whether the total number of similarity classes over all N is finite. They establish finiteness of every individual squarefree class and a global weighted zero-density statement. No BSD, GRH, or finiteness assumption on Shafarevich–Tate groups is used.

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